

Feedback, Reputation and Status in a Resistant Social Reward Economy

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1 Introduction

1.1 Motivation and gap

The *Social Reward Economies* paper describes resistance as an alternative reward economy but leaves feedback, reputation and status loosely specified with feedback quantities used in the inequalities measuring the effectiveness of resistance. This report provides these three definitions.

1.2 Scope

This report defines three quantities and how they express features of resistance in the #MeToo example. It does not define a decision rule, success conditions for resistance and other key parts of implementing resistance in a multi-agent system. The reason for this is that these depend on the definitions of the three stated quantities and should not be considered until these definitions are finalized.

2 The Example

2.1 #MeToo

An example is the formation of the #MeToo Movement. There are three types of people in this scenario: those in charge in the dominant group (such as high ranking corporate officials) the members of the #MeToo movement and finally the general public (which is just everyone else in society). In this setup it is clear that those in charge benefit at the expense of many women in the workplace and by extension many members of the #MeToo movement. Hence, the group in charge will have an incentive to preserve the dominant norm, which advocates for not calling out sexual harassment, as this directly benefits them. Conversely, women and the #MeToo movement wish to overturn the dominant norm in favour of their resistant one, that sexual harassment should be made public and the perpetrators should be shamed, as this benefits them. However, the general public is not strongly affected by the behaviours of either of the above groups, yet they make up a majority of the population.

The pre-movement norm is that individuals should stay silent about sexual assault and the movement's counter-norm is that they should instead disclose these cases and hold the perpetrators accountable.

2.2 Schemas and states

There are two schemas: dominant σ_s (silence/excuse) and resistant σ_r (disclose/hold accountable). Let $S_r \subseteq S$ be the set of states where sexual harassment is at issue. Outside of S_r , utilities are roughly uniform.

2.3 Utilities

Type of agent j	$u_j(s, \sigma_s)$	$u_j(s, \sigma_r)$
G_d (in power)	$\gg 0$	< 0
G_n (public)	> 0	> 0
G_r (movement)	$\ll 0$	$\gg 0$

Table 1: Utilities for $s \in S_r$.

2.4 Contrast cases

The above example is only a specific case where the dominant group and the subgroup are in direct opposition. However, there are other cases. First, consider the example of the accessibility movement: the public and those in charge do not gain from the old norm. Or, consider fans of a sports club: this is a subgroup with its own norm and network, unaffected by and not affecting the rest of society.

3 Minimal Setting

- States $s \in S$ with probabilities $p(s)$; actions $A(s)$.
- Utilities $u_j(s, a) \in \mathbb{R}$, heterogeneous across agents, positive or negative.
- Agent k 's behaviour as a distribution over actions, $O_k(s)$: the softmax policy from the *Learning Norms* paper. This differs from the SRE paper's $O_i(s)$, which denotes a schema and can be viewed as a proxy for k 's actual schema.
- Groups: a larger group G and a subgroup $G_r \subset G$. A generic group is written $H \in \{G, G_r\}$.
- Participation: rate $\mu_{j,p}^H$ and activity probability $\theta(\mu) = 1 - e^{-\mu}$, used only to define audience weights.

4 Definitions

4.1 Social feedback

Definition 4.1 (Social feedback). For agents $j \neq k$ and a state s , the feedback from j to k is j 's expected benefit from k 's behaviour:

$$f_{jk}(s) = \mathbb{E}_{a \sim O_k(s)} [u_j(s, a)]. \quad (1)$$

Here it is crucial that $u_j(s, a)$ can be both positive and negative so that social feedback can be both positive and negative. This is necessary as in the #MeToo example a person in power gives negative feedback to a survivor who discloses but the movement members give this action positive feedback.

4.2 Audience weights

Definition 4.2 (Audience weights). For a group H , let

$$\nu_j^H = \frac{\theta(\mu_{j,p}^H)}{\sum_{l \in H} \theta(\mu_{l,p}^H)}, \quad j \in H. \quad (2)$$

This normalization is important so that we get a probability distribution over H which expresses the probability of any given agent participating. This is important as it relates to how much other agents care about j 's feedback. If j does not participate regularly then their feedback is less valuable.

4.3 Reputation

Definition 4.3 (Reputation). For a group H and a member $k \in H$,

$$R_k^H(s) = \sum_{j \in H \setminus \{k\}} \nu_j^H f_{jk}(s). \quad (3)$$

Reputation is the audience-averaged feedback within H . It is important that there are separate reputation structures in G and G_r as otherwise there could not be any counter-public. In the #MeToo example, an advocate of the movement has high R^{G_r} and a much lower, possibly negative R^G while a person in power has high R^G .

4.4 Status

Definition 4.4 (Status). For $k \in H$, let $w_j^H = \nu_j^H [R_j^H]_+$ be rater j 's weight, where $[x]_+ = \max(x, 0)$, and set

$$S_k^H(s) = \sum_{j \in H \setminus \{k\}} w_j^H f_{jk}(s). \quad (4)$$

Status is reputation-weighted feedback: feedback from well-regarded raters counts more. The reputation term in the feedback weight is clipped at zero as a discredited rater should be ignored not have their feedback sign-flipped. In the #MeToo example, an early discloser receives negative feedback from powerful members of G but strongly positive feedback from respected movement members. This is the support that makes speaking out survivable.

Remark 4.1. This is not the status defined in the *Learning Norms* paper, where S_k sums each follower's reputation $R_i(\pi_k)$, i.e. the benefit to everyone else. This status sums raters' own perceived benefit; it recovers that definition only if the weights are follower indicators and f is replaced by R_i . This setup also has a hierarchy of influence among the defined quantities: weights depend on reputation and status depends on weights.

5 How the Three Quantities Pertain to Resistance

In this section we describe how the three quantities pertain to resistance. Specifically, which pattern across feedback, reputation and status distinguish a resistant subgroup from an ordinary one.

Agent (behaviour)	R^G	S^G	Feedback from G_r
Person in power (σ_s)	Very Positive	Very Positive	Very Negative
Advocate who discloses (σ_r)	Very Negative	Very Negative	Very Positive
Bystander (σ_s)	Positive	Positive	Negative

Table 2: Standing in G for representative agents.

5.1 The configuration before the movement

5.2 An alternative economy: the pattern that signals resistance

For G_r to have its own economy it must have a network separate from G where status, reputation and social feedback can be established for members of the subgroup based only on the utilities of the subgroup. The contrast between the values of these quantities in the subgroup vs total group are illustrated below:

Agent (behaviour)	R^{G_r}, S^{G_r} (movement's view)	R^G, S^G (larger group's view)
Advocate who discloses (σ_r)	Positive	Negative
Movement leader (σ_r)	Very Positive	Very Negative

Table 3: The same agents evaluated by the two audiences.

Remark 5.1. This is a description of what an alternative economy looks like and the paper has the equations

$$\left| \sum_{i \in G_r} w_i f_i^r(\sigma_r, s) \right| \gg \left| \sum_{j \in G \setminus G_r} w_j f_j(\sigma_r, s) \right|$$

$$\left| \sum_{i \in G_r} w_i f_i^r(\sigma_s, s) \right| \gg \left| \sum_{j \in G \setminus G_r} w_j f_j(\sigma_s, s) \right|$$

as the necessary conditions for successful resistance to take place and these are stated in terms of weighted feedback as defined above. However, how such feedback determines whether the subgroup's norm spreads is not discussed and is a next step once the above definitions are solidified.

5.3 The cost of speaking out

A discloser of sexual assault in the #MeToo example receives negative feedback and has low status in the larger group but positive feedback and high status inside G_r . If $S^{G_r} \gg S^G$ then speaking out is costly as the negative outcome in the larger group outweighs the positive one in the subgroup. Conversely, if the inequality is reversed then speaking out is beneficial and this mechanism is what should sustain the subgroup and promote resistance.

5.4 Contrast cases

Considering the alternate subgroup examples from before, they can also be describe by feedback, reputation and status only with different values:

Example	Feedback from the wider group to the subgroup's norm	Reputation and status pattern
#MeToo	Negative	negative
Accessibility	Positive	Slightly Positive
Sports fans	Neutral	Neutral

Table 4: Pattern of the three quantities in each example.

6 Discussion

6.1 Relation to Learning Common Norms

The status defined here is similar to that defined in the *Learning Norms* paper except now utilities are signed, but the overall structure (use of utilities and softmax policy) are largely the same.

6.2 What is deliberately left out

There were multiple points not included in this report: how feedback aggregates into a choice between schema; how much attention agents give each group; formal conditions for a subgroup norm to spread or to displace the dominant one. All of these questions are the natural next steps to consider once the above three definitions are settled.

6.3 Limitations

- The #MeToo example is used to motivate the definitions, so agreement with the example is plausibility, not validation.
- The definitions are static and say nothing about whether the described patterns arise.